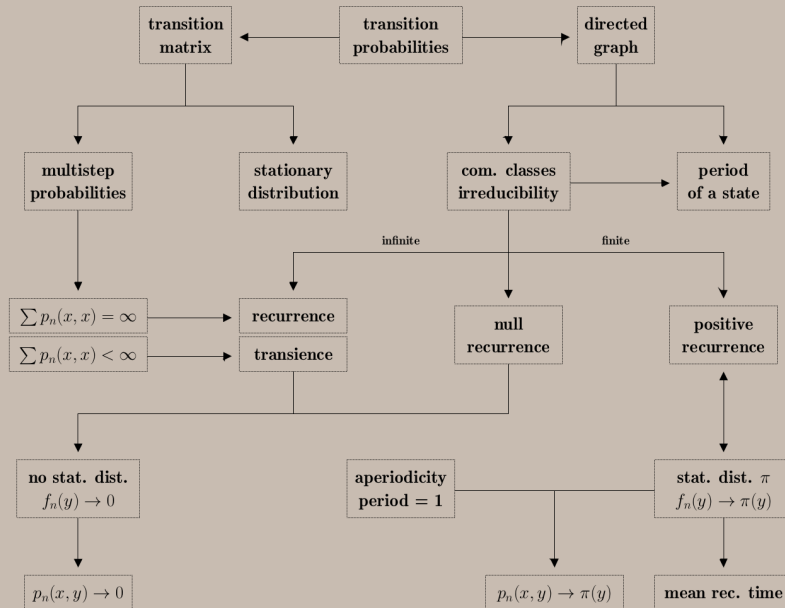
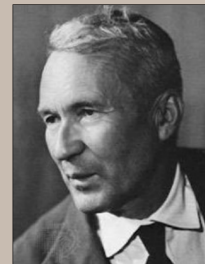
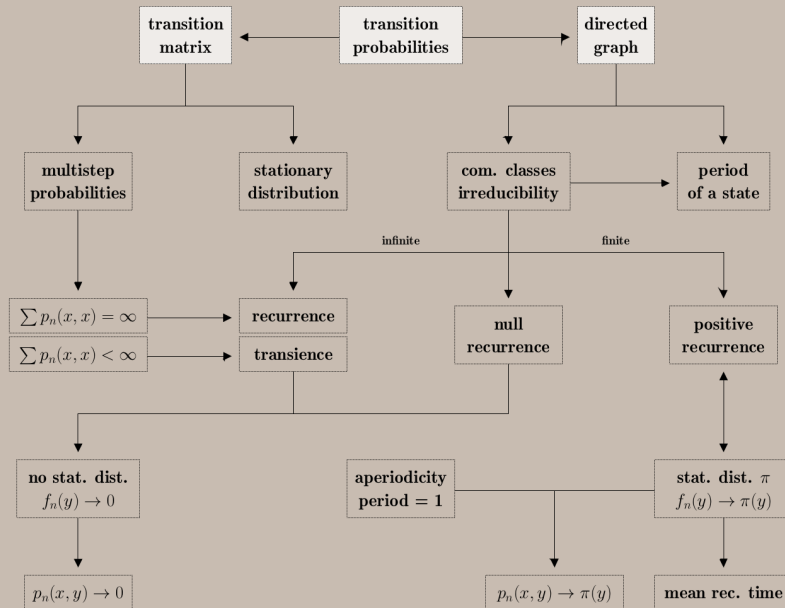




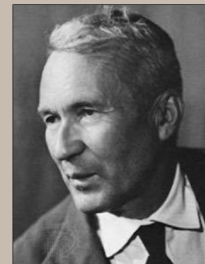
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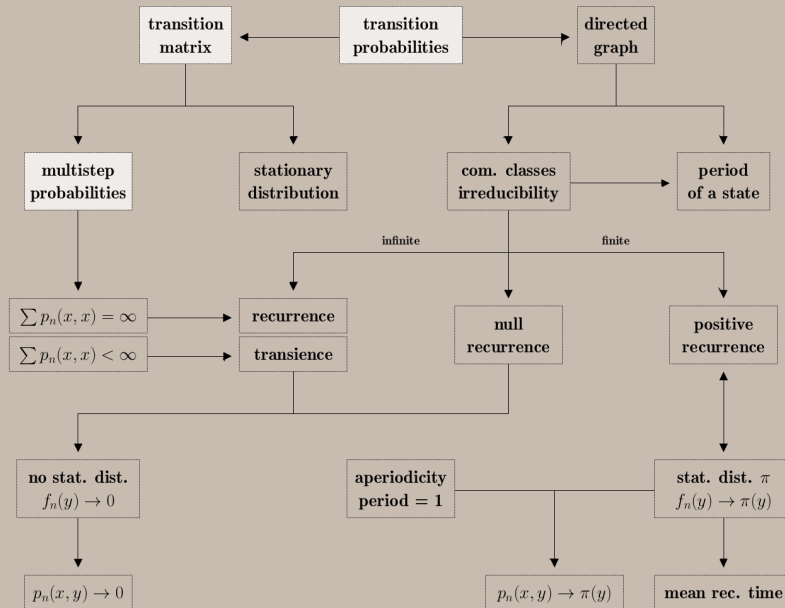
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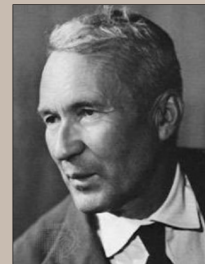
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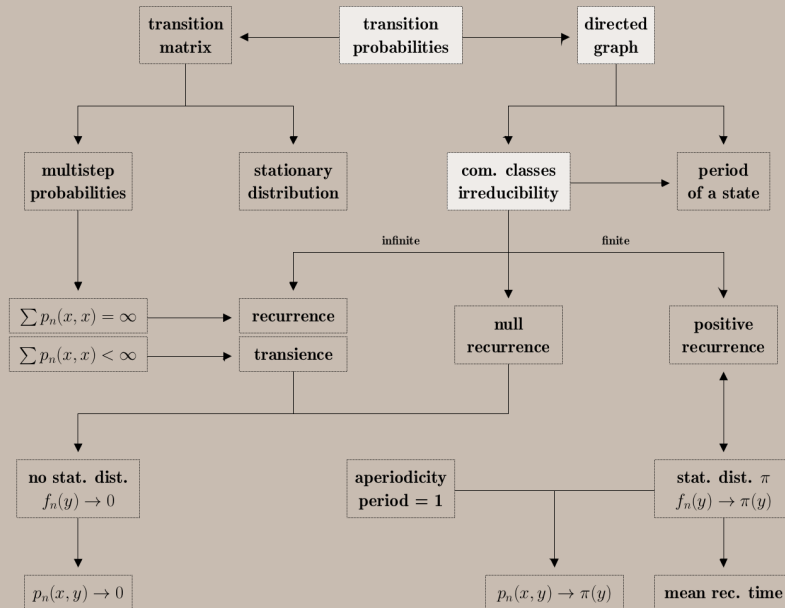
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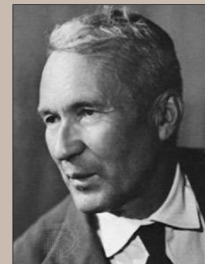
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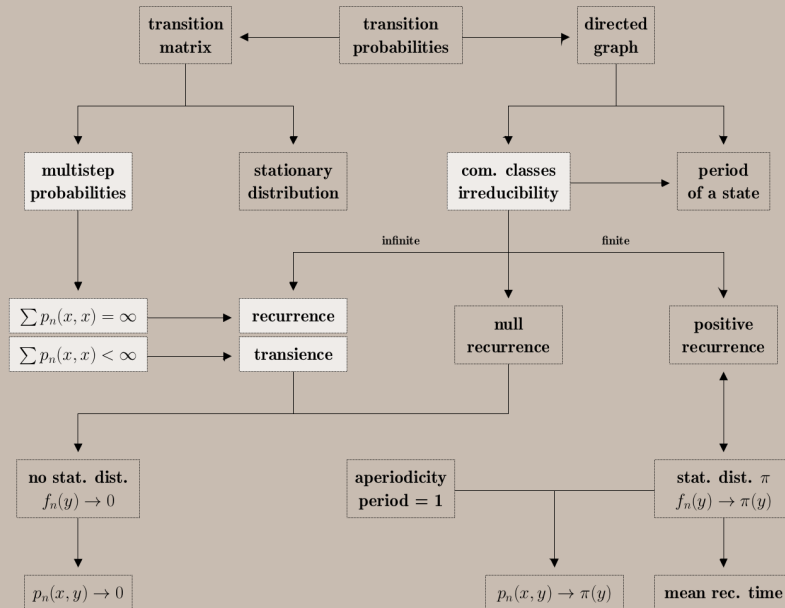
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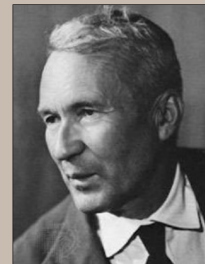
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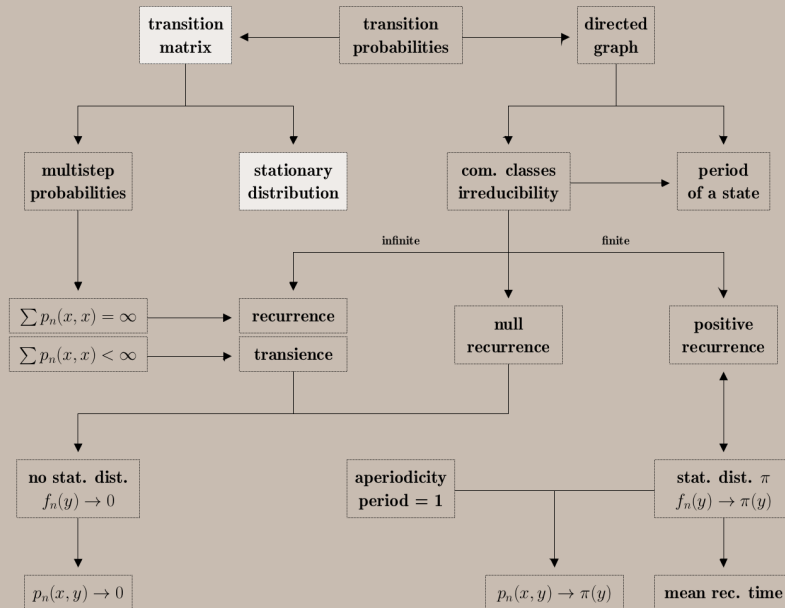
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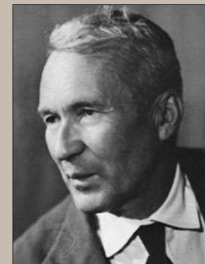
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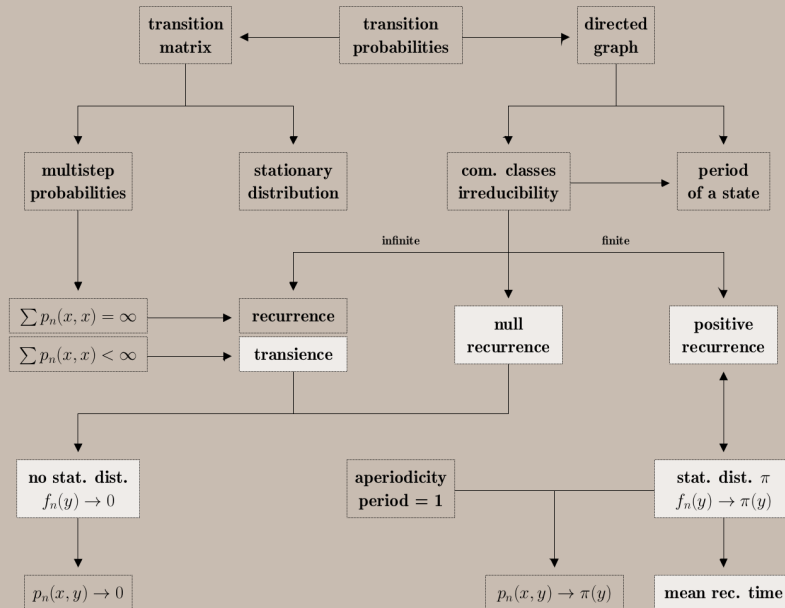
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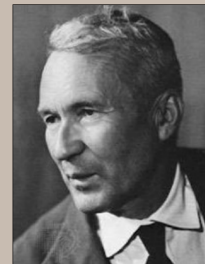
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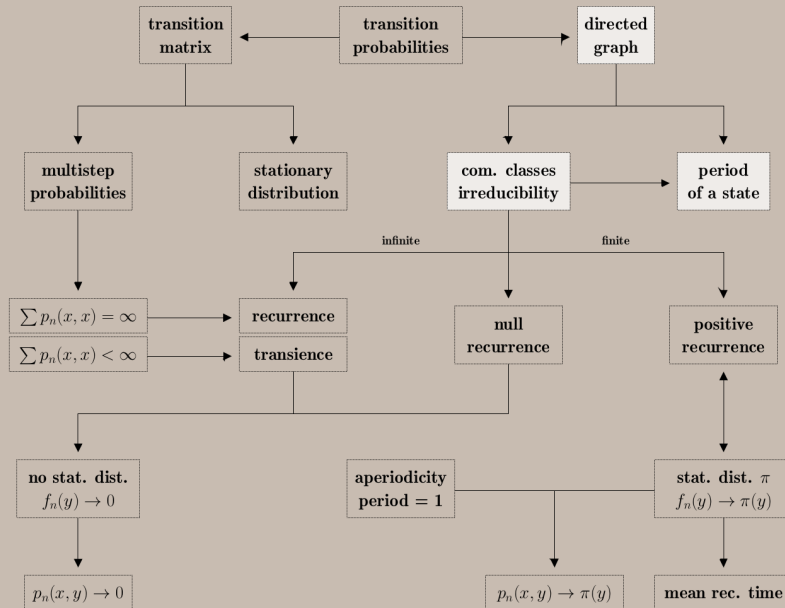
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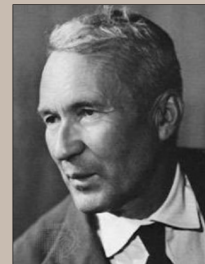
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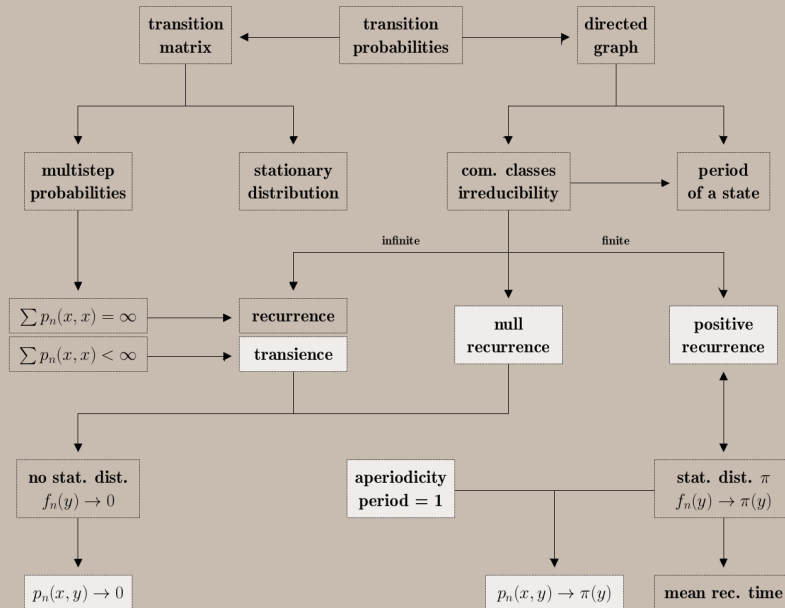
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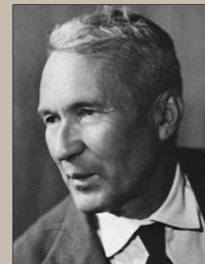
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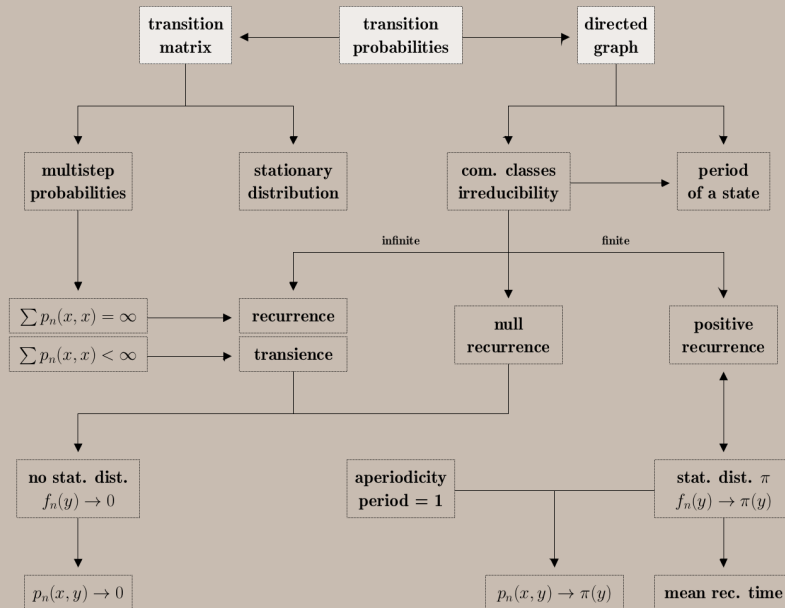
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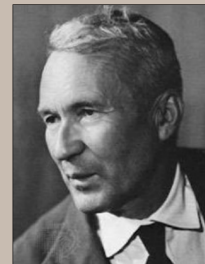
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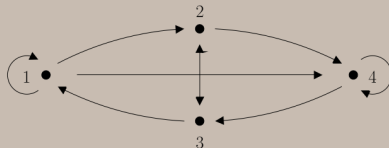
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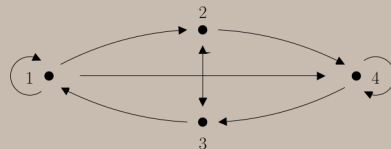
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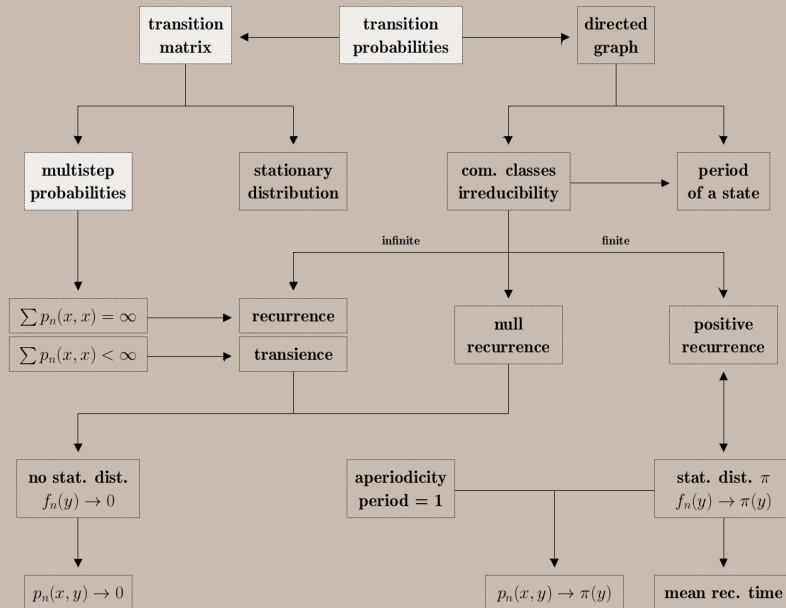
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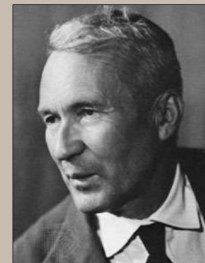
quantitative aspects:
multistep probabilities and stationary distribution



qualitative aspects:
communication classes and period



Andrey Markov



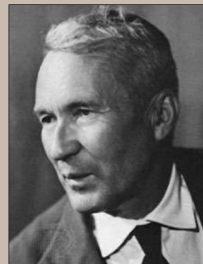
Andrey Kolmogorov

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Sydney Chapman



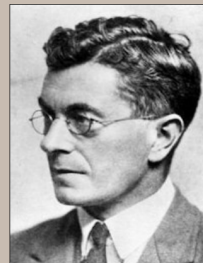
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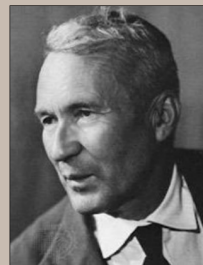
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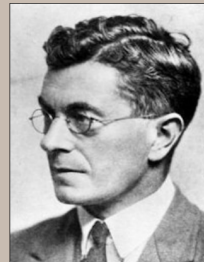
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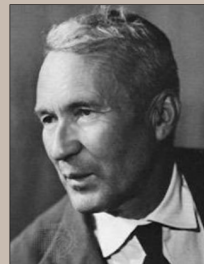
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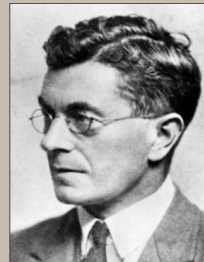
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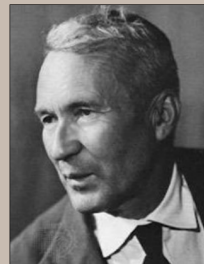
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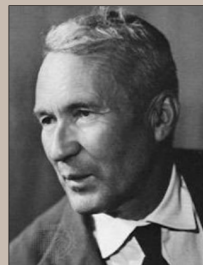
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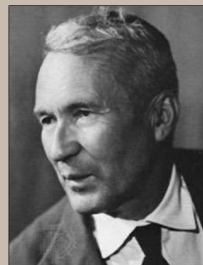
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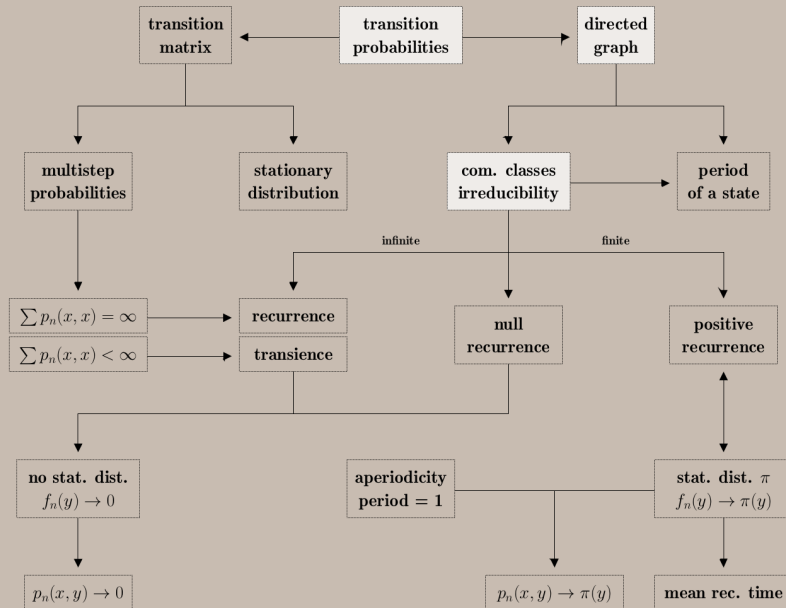
Note: the matrix P^n is typically difficult to compute in practice but the lemma is useful to develop the theory (stationary distribution).



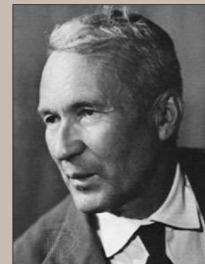
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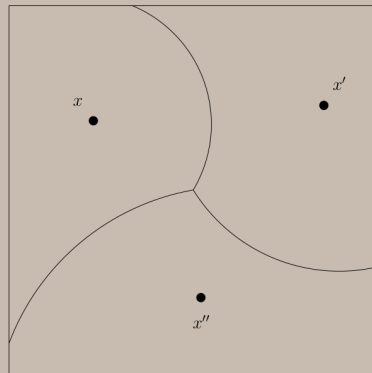
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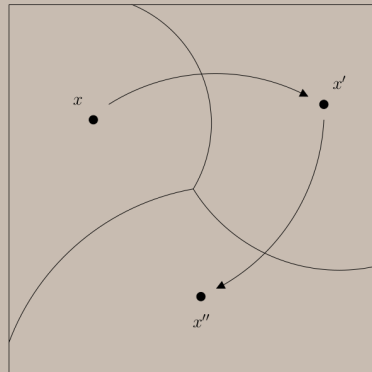


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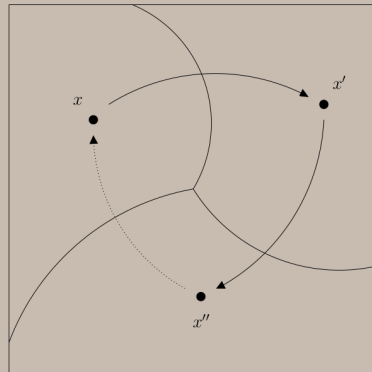


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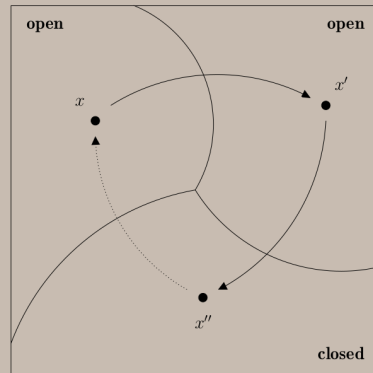


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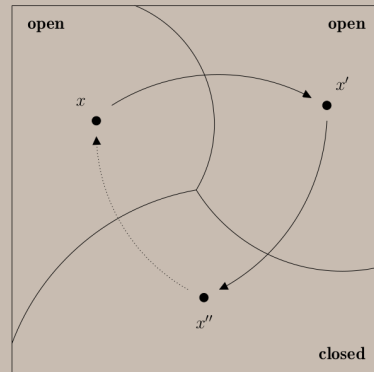
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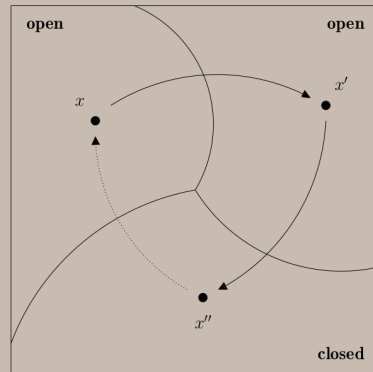
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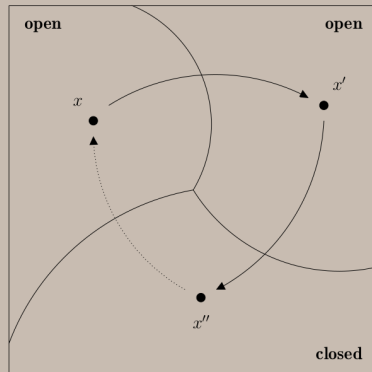
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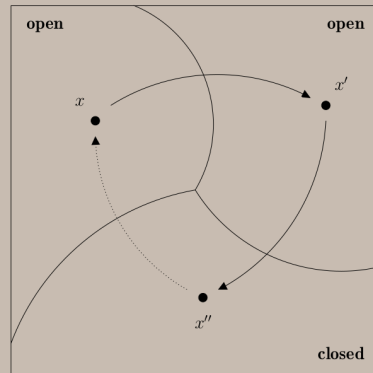
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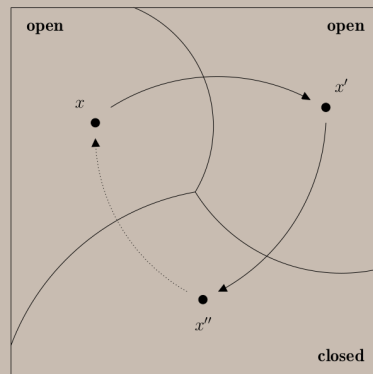
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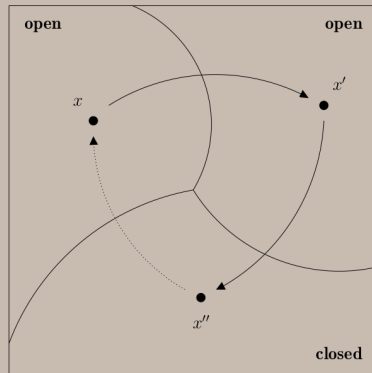
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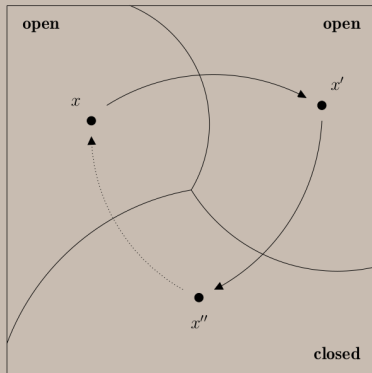
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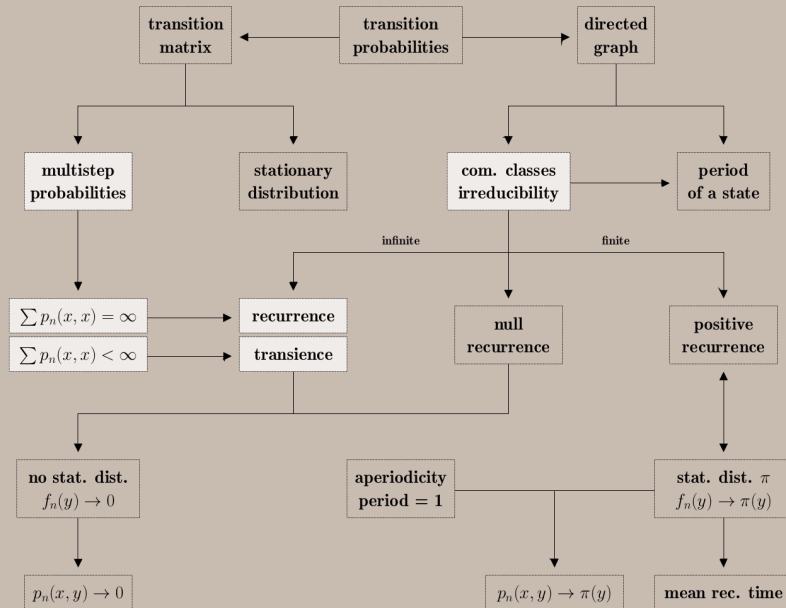
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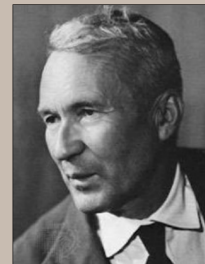
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Andrey Markov



Andrey Kolmogorov

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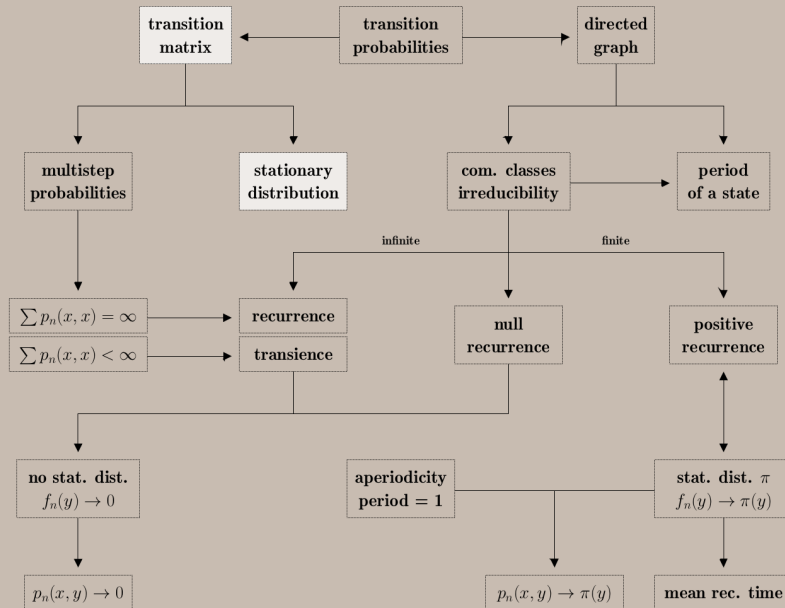
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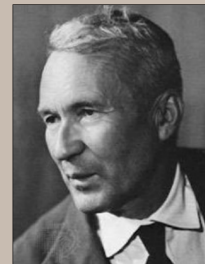
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Andrey Markov



Andrey Kolmogorov

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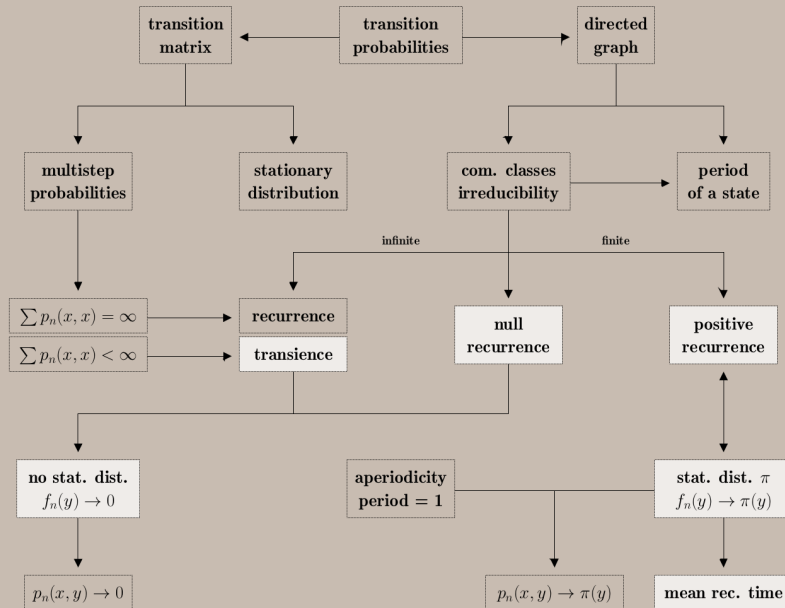
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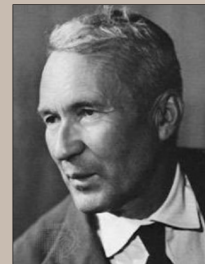
$$(1, 1, \dots, 1) P = (1, 1, \dots, 1) \implies \pi = \text{Uniform}(S) \text{ is stationary.}$$

Reversible distribution: $\pi(x) p(x, y) = \pi(y) p(y, x)$ for all $x, y \in S$ (detailed balance condition).

$$(\pi P)(x) = \sum_{y \in S} \pi(y) p(y, x) \stackrel{\text{DBC}}{=} \sum_{y \in S} \pi(x) p(x, y) = \pi(x) \implies \pi \text{ is stationary.}$$



Andrey Markov



Andrey Kolmogorov

Fraction of time spent in a state.

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Theorem. Positive recurrence and null recurrence are class properties.

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Theorem. Assume irreducibility. Then,

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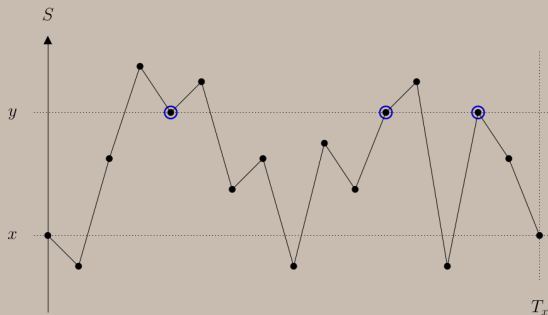
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1) Cycle trick:

$$\mu_x(y) = \sum_n P_x(X_n = y, T_x > n)$$

expected number of
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 &= \sum_n P_x(X_{n+1} = z, T_x > n+1) \\
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$$\begin{aligned} \mu_x(S) &= \sum_{y \in S} \mu_x(y) = \sum_{y \in S} E(\# \text{ visits in } y \text{ between two visits in } x) \\ &\stackrel{\text{MCT}}{=} E(\# \text{ visits in any state between two visits in } x) \\ &= E_x(T_x) < \infty. \end{aligned}$$

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$$(1) + (2) \implies f_n(y) \sim m/T_y^m \xrightarrow{a.s.} 1/E_y(T_y) \quad \text{as } m, n \rightarrow \infty.$$

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Dominated convergence theorem $\implies$

$$\pi(y) = \lim_{n \rightarrow \infty} E_\pi(f_n(y)) \stackrel{\text{DCT}}{=} E_\pi(\lim_{n \rightarrow \infty} f_n(y)) = E_\pi(1/E_y(T_y)) = 1/E_y(T_y).$$

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Irreducibility $\implies p_n(x, y) > 0$ for some $n \geq 0 \xRightarrow{\text{THM}}$

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Theorem. **Irreducibility** + **stationary distribution** $\pi \implies$ **positive recurrence** of all states.

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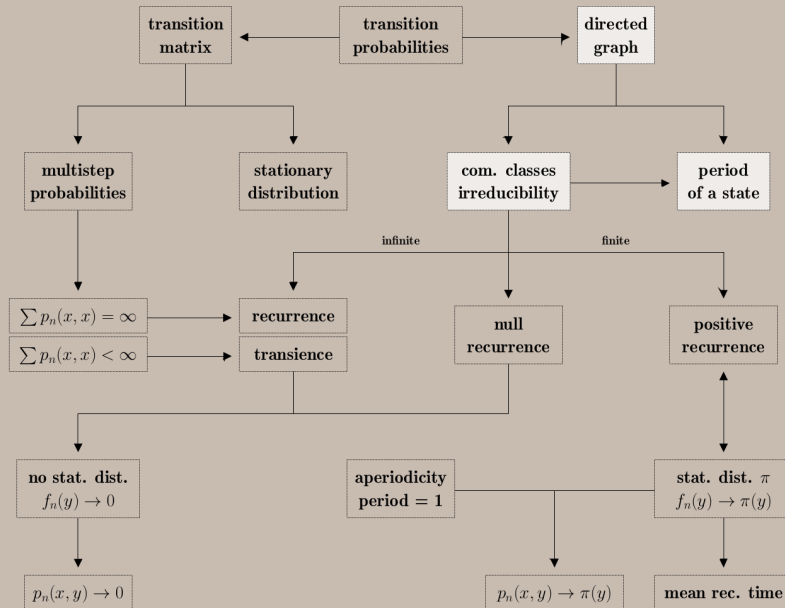
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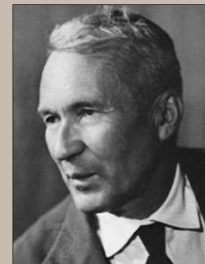
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Andrey Markov



Andrey Kolmogorov

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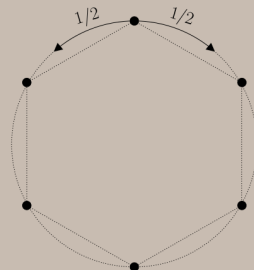
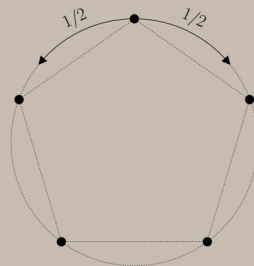
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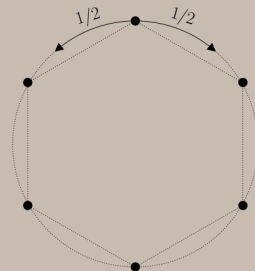
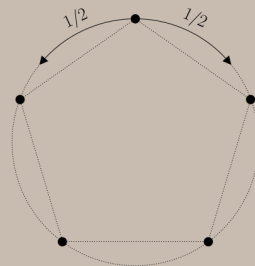
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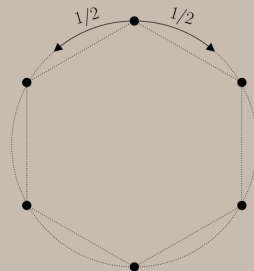
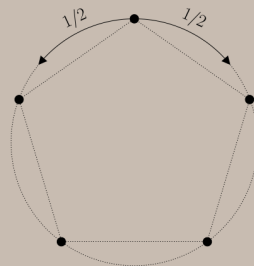
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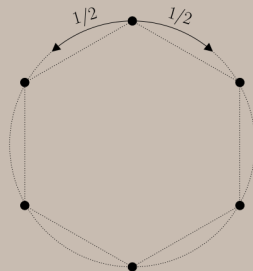
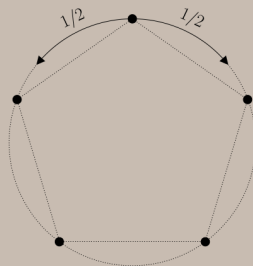
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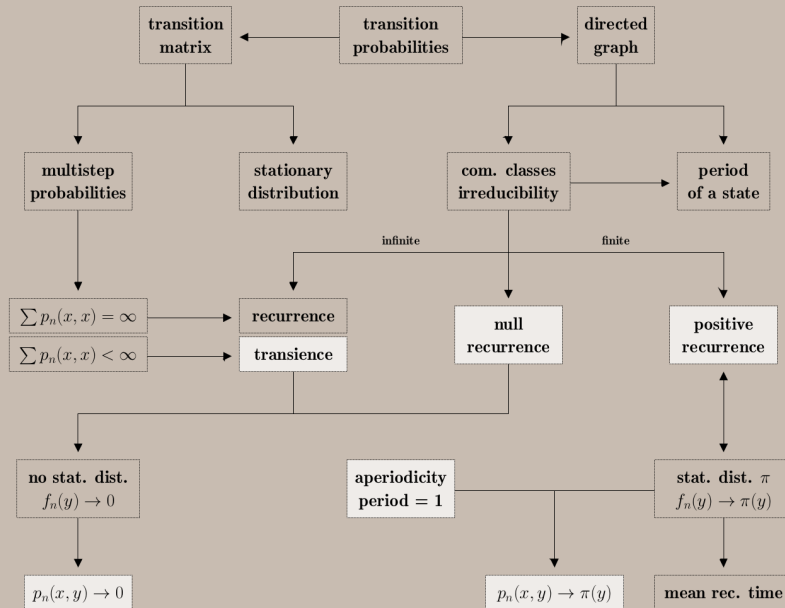
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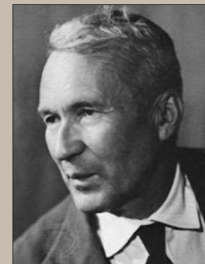
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$$p_{2n+1}(x, x) = 0 \text{ and } p_{2n}(x, x) \rightarrow 1/3.$$





Andrey Markov



Andrey Kolmogorov

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Let $k \geq K = c^2$. **Euclidean division** $\implies$ there exist $q, r \in \mathbb{N}$ with $r < c$ such that

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for all k large $\implies Z_n$ is irreducible.

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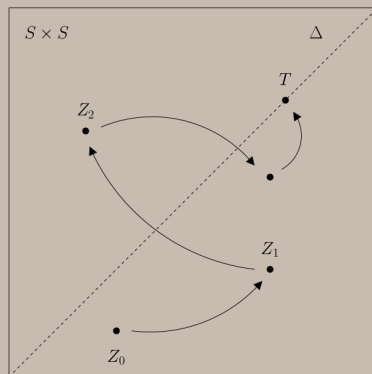
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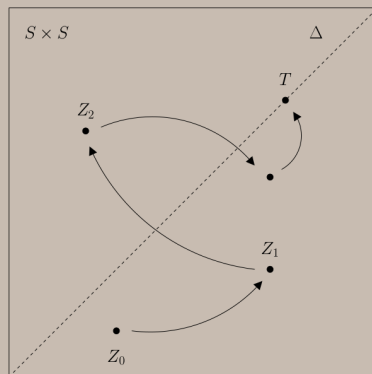
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After time T , $X_n = Y_n$ in distribution $\implies$

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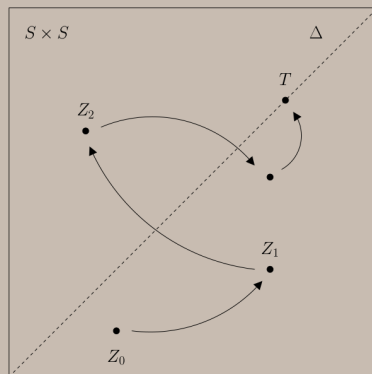
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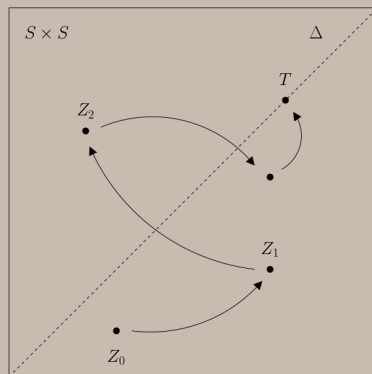
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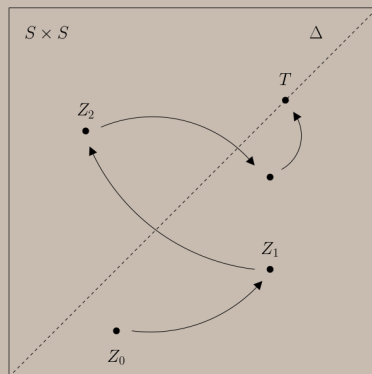
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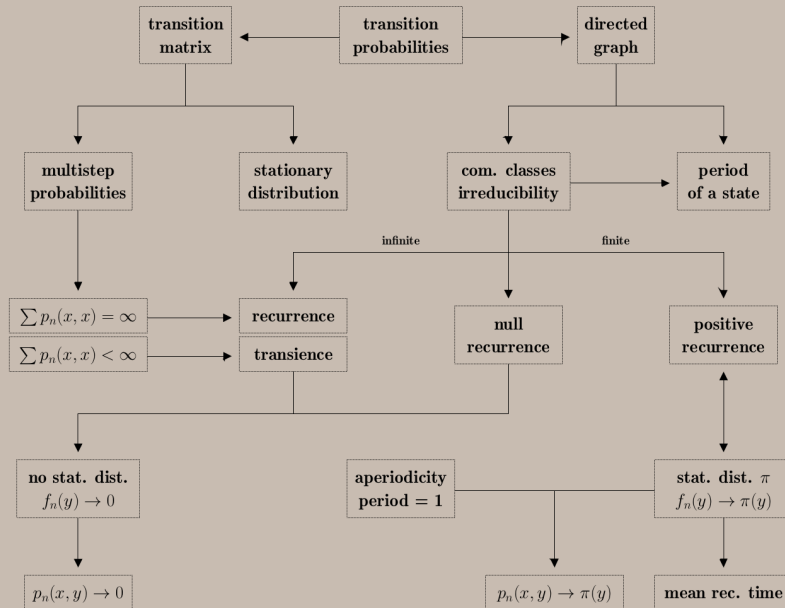
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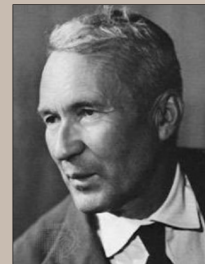
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